Network Agent Operational Lifecycle: Network Agent Observability, Intervention and Control

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IETF ICON Side Meeting

Wednesday 22 July 2026

8:00 - 9:30 (Vienna Time)

Park Suite 4

Background and Recap

- AINETOPS github was created (https://github.com/IETF-OPS-AD/AINETOPS) to
  - Coordinate or provide classification for AI related work distributed in different WGs
  - Keep track of BOF, Side meetings, Mailing lists, Hackathon Projects
- AI for Network Operation wikipage (https://wiki.ietf.org/e/en/group/ops/aiops) provides guideline for AI driven Network Operation work
- AI topics can be break down into Network for AI and AI for Network
  - Network for AI such as: AI Network Multicast, AIDC, FANN
  - AI for Network such as: DAWN, AgenticOPS side meeting
- AI Topics are not new
  - AIPref WG
    - Vocabulary for expressing AI-related preferences
    - Attaching preferences with content in IETF-defined protocols and formats
  - Webbotauth WG
    - Cryptographically authenticating method for AI Crawlers
  - NMOP WG
    - Network Anomaly architecture, Lifecycle and Metadata Semantics
    - Network Incident YANG
  - NMRG
    - Research challenges in Coupling AI and Network Management
    - AI Deployment Consideration
    - Agentic AI Architecture and Problem Statement
    - …

Why are we here?

- We always believe AI for good
  - UC Cases: service complaint Handling, IP+Optical cross layer cross domain fault management, etc.
  - Value:OPEX/CAPEX reduction,ROIservice (Service Manager), site visit reduction(Field Engineer), etc.

- What if AI for bad/Network AI Agent goes wrong?
  - **Non-Deterministic Execution Paths**
    - same input with different chains of tool calls, retries, and reasoning steps
  - **Cascading Failures That Look Like Success**
    - An agent can return a confident-sounding wrong answer after silently failing to retrieve context
  - **Hidden Cost Accumulation**
    - single user query might trigger 15 LLM calls across different models
    - agent loop can burn through a monthly token budget in hours
  - **Extreme Latency Variance**
    - Same query might complete in 800ms or 30 seconds depending on the reasoning path

Why are We here

- What can we do about it if Network Agent goes wrong
  - Can We See it? -> Agent Observability
  - Can we set boundary to prevent agent from violating policies? Agent Control
  - Can we stop high risk operation? -> Human Oversight

- Goal of this side meeting:
  - Follow up the previous AgenticOPS side meeting
    - https://github.com/danielkinguk/AgenticOps
  - Explore key areas where standardization is required to support the design, implementation, and operation of Network Management Agent Observability, Intervention and Control

Meeting Agenda

- Opening Remarks(IETF meeting rules, agenda arrangement, and objective) - 5 min Moderators
- Network Agent Observability, Intervention and Control (ICON) Problem Statement -10 min + 5min Qin Wu
  - Reference: https://datatracker.ietf.org/doc/draft-wnd-opsawg-icon-ps/
- Network Agent Observability, Intervention and Control (ICON) Architecture and Requirements 10 min +5 min Qiufang Ma
  - Reference: https://datatracker.ietf.org/doc/draft-mcw-opsawg-icon-requirements/
- Current work on Observability, Traceability at AAIF 10min + 5min Daniel King
- Open Discussion on this new proposed Initiative 40 min All
- Wrap-up / Next steps 5 min Moderators

Discussion

- Is the problem Real? What else is missing?
- Is IETF right place to develop solution to address this problem?
- Who are interested to contribute?